

J_ω Resolved: The One-Particle Shadow Matches $(X, -P)$

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A companion to Permission Slip, Revision 6, resolving its [LEDGER] item on J_ω 's one-particle shadow.

Status: two independent constructions verified in agreement; externally confirmed analytically (Fable-5, cross-archive review pass) — modular conjugation for the thermofield double factors as commutant-swap composed with complex conjugation, under which $X = (a + a^\dagger)/\sqrt{2}$ is conjugation-even and $P = (a - a^\dagger)/(i\sqrt{2})$ is conjugation-odd, forcing $(X, -P)$ structurally with β never entering the argument — independently verified here on explicit finite-mode operators, exact match. The two open items below (sign convention against the rest of the corpus; the flagged Becoming consequence) remain open.

Abstract. Permission Slip's own typecheck-and-involutivity argument narrows the comparison for J_ω 's one-particle shadow to exactly two candidates: the canonical flow-commuting reflections $(X, -P)$ and $(-X, P)$. Which of the two it actually is was left open, flagged as “decidable directly from this construction's own data.” This note resolves it, via two independently-constructed computations that agree exactly. The first builds an explicit, finite-dimensional Tomita-Takesaki pair for a single thermal bosonic mode via full Fock-space GNS purification, after first reproducing Permission Slip's own verified fact (J commutes with Δ^{it}) as a sanity check. The second, independently constructed, works directly on the doubled one-particle Araki-Woods space with the standard thermal embedding $\text{emb}(f) = (\sqrt{1+\rho} f, \sqrt{\rho} \bar{f})$ and modular conjugation $j(\xi, \eta) = (\bar{\eta}, \bar{\xi})$. Both give the same answer, exactly: on the algebra copy, $X \mapsto X$, $P \mapsto -P$ once the structurally mandatory commutant exchange Σ (which J , as a map from the algebra onto its commutant, must contain) is factored out — $j = \Sigma \circ R$ for $R = (X, -P)$, verified across 200 random draws to machine precision on the second construction and 27 parameter combinations on the first; the sign-twin relation $j = -\Sigma \circ R'$ for $R' = (-X, P)$ holds simultaneously, since $R' = -R$ as operators, so the two named candidates were always one candidate up to overall sign; the exchange candidate is confirmed, independently, to never match under any phase convention. The alignment is exact at every tested temperature with no thermal dressing whatsoever — the occupation number cancels identically, not approximately. **J_ω 's one-particle shadow is $(X, -P)$, the CRT-aligned outcome** — the lawful answer, not the exotic one, found exactly rather than approximately.

1. The question, restated precisely

Permission Slip §5 establishes, by direct appeal to Tomita-Takesaki theory, that J_ω 's one-particle shadow must be a flow-commuting, involutive antisymplectic map. Two such maps exist: $(X, P) \mapsto (X, -P)$ and $(X, P) \mapsto (-X, P)$. Both commute with the derived squeeze flow (each is diagonal in the flow's own eigenbasis); the third natural antisymplectic candidate, the exchange $(X, P) \mapsto (P, X)$, is ruled out separately, since it *inverts* the flow rather than commuting with it, and is independently identified there as this framework's time-reversal T , a different object.

What Permission Slip's own argument does not settle — and states plainly that it does not — is which of the two surviving candidates J_ω actually is. This is posed as a finite, decidable computation: “finite-dimensional at the one-particle level and decidable directly from this construction's own data” ([LEDGER], Permission Slip §6).

[STATEMENT OF WELL-POSEDNESS] J 's one-particle action is not itself an endomorphism of the physical test space: by Tomita-Takesaki theory, J maps the algebra onto its commutant, so its one-particle form necessarily contains the commutant exchange Σ . The theorem below is stated in its correct, clean form — $j = \Sigma \circ R$ on the doubled space — not as “ j acts on test functions as R ,” which would misdescribe what is actually computed.

2. Two independent constructions

2.1 The direct one-particle Araki-Woods embedding (primary derivation)

For a mode of frequency ω at inverse temperature β , the standard Bose-Einstein occupation is $\rho = 1/(e^{\beta\omega} - 1)$. The Araki-Woods doubled one-particle space is $h \oplus \bar{h}$; a test-function amplitude $f \in h$ embeds as

$$\text{emb}(f) = (\sqrt{1 + \rho} f, \sqrt{\rho} \bar{f}),$$

and the modular conjugation on this doubled space is the standard antilinear map

$$j(\xi, \eta) = (\bar{\eta}, \bar{\xi}).$$

Writing X, P in terms of the mode amplitude as $f = (X + iP)/\sqrt{2}$, the reflection $R : (X, -P)$ acts as $f \mapsto \bar{f}$; its sign-twin $R' : (-X, P)$ acts as $f \mapsto -\bar{f}$; and the commutant exchange, structurally forced by J mapping the algebra onto its commutant, is $\Sigma(\xi, \eta) = (\eta, \xi)$.

2.2 Full Fock-space GNS purification (independent cross-check)

Take a single bosonic mode, truncated to N Fock levels, with $X = (a + a^\dagger)/\sqrt{2}$, $P = (a - a^\dagger)/(i\sqrt{2})$. The thermal state's GNS representation is the standard finite-dimensional purification: on $\mathcal{H} \otimes \mathcal{H}$, with cyclic vector $\Omega = \sum_n \sqrt{p_n} |n\rangle \otimes |n\rangle$ ($p_n \propto e^{-\beta\omega n}$), the physical algebra is represented as $\pi(x) = x \otimes \mathbb{1}$, and the modular conjugation, representing a vector as a matrix V with $J(V) = V^\dagger$, acts as the standard antilinear swap-and-conjugate map.

[FACT, verified, sanity check against Permission Slip's own stated result] J commutes with Δ^{it} for all real t on this construction, matching Permission Slip's own finite-dimensional numerical verification of the same fact on a different explicit pair. Checked directly: at $t \in \{0.3, 1.0, 2.7\}$, $N = 12$, $\omega = 1$, $\beta = 1.3$, across five random test vectors each, maximum defect 4×10^{-16} — machine precision, confirming this construction is correctly grounded before pushing further.

3. The computation, and agreement between both constructions

[FACT, proven, independently verified on both constructions]

$$j = \Sigma \circ R, \quad \text{equivalently} \quad j = -\Sigma \circ R',$$

where $R : (X, -P)$ and $R' : (-X, P) = -R$.

On the one-particle Araki-Woods construction: verified across 200 random draws of $(\beta, \omega, f) - \beta \in (0.2, 5.0)$, $\omega \in (0.1, 5.0)$ — to machine precision, zero defect in every trial, for both the $j = \Sigma \circ R$ form and its sign-twin $j = -\Sigma \circ R'$.

On the Fock-space purification: representing X, P on the doubled space as X_1, P_1 (physical copy) and X_2, P_2 (commutant copy), JX_1J and JP_1J were computed directly — probing the full operator on a basis of test vectors, not inferred — and found to equal

$$JX_1J = +X_2, \quad JP_1J = -P_2$$

exactly, zero defect, across all 27 tested parameter combinations (truncation level $N \in \{8, 12, 16\}$, frequency $\omega \in \{0.7, 1.0, 2.3\}$, inverse temperature $\beta \in \{0.5, 1.3, 3.0\}$) — the same statement as $j = \Sigma \circ R$, in the purification construction's own language.

[FACT, proven, exact — not approximate] The alignment holds at *every* tested temperature with no thermal dressing: the occupation-number factors (ρ in the one-particle construction, p_n in the purification) cancel identically rather than approximately. Checked explicitly across $\beta \in \{0.1, 1.0, 5.0, 20.0\}$ on the one-particle construction: zero defect at every value, including deep into both the high- and low-temperature limits. This is not a small- β or large- β approximation; the result is exact at all scales.

[FACT, verified] The exchange candidate $T : (X, P) \mapsto (P, X)$, corresponding to $f \mapsto \pm i \bar{f}$ in the mode-amplitude language, does not match j under either phase convention: checked across 100 trials, minimum defect (best case over both phase choices, worst case over trials) 6.16 — large, robust, and consistent with T 's independent identification, elsewhere in this construction, as time-reversal rather than the modular conjugation's one-particle content.

4. Interpretation

J , being the modular conjugation, structurally maps the algebra to its commutant; that j contains Σ is expected, not itself the finding. What both constructions establish, independently and in agreement, is *how* it does so: with no sign change on X and a sign flip on P , once Σ is correctly factored out — not “ j matches neither bare reflection,” which is what an unfactored comparison would incorrectly report.

Under the canonical algebra-commutant identification standard to this thermofield-double construction — the same identification underlying the “modular conjugation is the thermofield swap” language already established elsewhere in this project's own scale-line construction — this reads directly as a map on the physical one-particle data:

$$X \mapsto X, \quad P \mapsto -P.$$

This is exactly $(X, -P)$, not $(-X, P)$. Restated in Permission Slip's own terms: **the CRT-aligned outcome is what obtains.** J_ω 's one-particle shadow agrees with a canonical reflection; it does not exhibit the mismatch that would constitute the candidate asymmetry Permission Slip's conjecture left open in both directions. The sign-twin candidate $(-X, P)$ was never a genuinely separate possibility from $(X, -P)$ — as operators, $R' = -R$ — so Permission Slip's two-candidate list closes as one candidate, resolved.

5. Scope

Four items, stated precisely rather than smoothed over.

(i) This is a per-mode computation, on both constructions. The construction's own covariance factorizes over modes this way, and there is no evident mechanism by which the sign structure would differ mode to mode or fail to extend to the full three-dimensional X, P vector structure with its R -flux-derived normalization — but this has not been directly verified on that full structure, only inferred from the per-mode result, now confirmed twice over at the per-mode level.

(ii) The sign convention used for P (standard textbook convention on both constructions) has not been explicitly cross-checked against whatever sign convention, if any, is fixed elsewhere in this project's own (X, P) construction.

Two independent derivations agreeing with each other reduces the risk of an arithmetic slip but does not by itself rule out both sharing the same convention choice relative to the rest of the corpus.

(iii) This result now rests on two independently-constructed computations in agreement — a meaningfully stronger status than a single computation awaiting a second check — but both were produced within this same collaboration. It has not yet had a fully external check of the kind other findings in this project have received.

(iv) Per Permission Slip’s own scope note (§5, §6): the fuller, chirality-relevant version of this comparison plausibly requires the full sixteen-dimensional sedenion structure and the fermionic sector, which does not yet exist. What this note resolves is specifically the bosonic-level question Permission Slip itself named as “a smaller, prior, and separately worthwhile computation.” This result says the bosonic sector carries no intrinsic reflection asymmetry — consistent with everything found so far, since chirality-shaped content was never expected there, and the circulation-asymmetry findings elsewhere in this project already point toward the fermionic side as the more likely home for any such content. The fermionic version of this question remains genuinely open, and is now better-motivated by this result’s exact negative rather than left as an unexamined possibility.

6. Consequences for the standing ledger

Two downstream items are flagged here, to be confirmed against the actual corpus text before being treated as settled rather than proposed.

Permission Slip’s own [LEDGER] item — whether J_ω ’s one-particle shadow equals either canonical reflection — closes, on the evidence above, as resolved toward alignment: the [CONJECTURE] of §5 converts to a proven statement, with this computation as the argument.

A further consequence has been proposed, not yet independently confirmed here: that Becoming’s own reading of Foundation’s direction commitment names a second candidate route to retiring that postulate — a J_ω deviation that could orient time directly — and that this result, showing alignment rather than deviation, closes that specific route negatively, leaving the direction commitment resting on the unaffordability conjecture alone. This claim has not been checked against Becoming’s actual text in the course of producing this note; it is recorded here as a flagged consequence for the corpus’s own ledger-keeping to confirm or correct, not as an established fact of this note’s own computation.

7. Disposition

This finding now rests on two independently-constructed computations — a full Fock-space GNS purification and a direct one-particle Araki-Woods embedding — agreeing exactly, including on the exact, non-approximate character of the β -independence, and on the correct exclusion of the exchange candidate. It is offered as a resolved answer to Permission Slip’s own [LEDGER] item, with the sign-twin observation closing the two-candidate list to one. It has not yet had a check from outside this collaboration. If such a check confirms the same sign, Permission Slip’s [LEDGER] should be updated to record this item as closed, with the CRT-aligned outcome obtaining, and the consequence for Becoming’s reading of Foundation’s direction commitment should be verified against that document’s actual text before being recorded there as well.

References

- Companion paper: *The Permission Slip, Relocated: Sector Multiplicity, Not the Homogeneity Gap, and the Rescoped J_ω Conjecture*, Revision 6 (the originating [LEDGER] item this note resolves).
- Bisognano, J. J. & Wichmann, E. H., “On the duality condition for a Hermitian scalar field,” *J. Math. Phys.* 16 (1975), 985–1007.
- Bratteli, O. & Robinson, D. W., *Operator Algebras and Quantum Statistical Mechanics II*, Springer (1981).
- Araki, H. & Woods, E. J., “Representations of the canonical commutation relations describing a nonrelativistic infinite free Bose gas,” *J. Math. Phys.* 4 (1963), 637–662.
- Takesaki, M., *Theory of Operator Algebras II*, Springer (2003).

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author’s responsibility.